

Cycles Refuted: A Pre-Registered Three-Gate Falsification of the Kitchin, Juglar, Kuznets and Kondratieff Cycles across Six Macroeconomic Panels, 1700–2024

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Abstract

We test whether the four canonical economic cycles—Kitchin (3–5 years), Juglar (7–11 years), Kuznets (15–25 years) and Kondratieff (40–60 years)—are statistically distinguishable from coloured noise across six heterogeneous macroeconomic panels spanning 1700–2024 (Bank of England Millennium, Jòrdà-Schularick-Taylor R6, Bank for International Settlements credit aggregates, World Bank annual indicators, FRED/Eurostat quarterly, and a sectoral panel). A pre-registered three-gate falsifiable protocol is applied to every cell of the $\{\text{cycle}\} \times \{\text{panel}\} \times \{\text{variable}\}$ grid: (i) Gate 1 requires the band-pass power in the canonical band to exceed both an AR(1)-bootstrap null and a phase-randomised null at $\alpha = 0.05$ with $B = 1000$ surrogates; (ii) Gate 2 requires consensus between at least three of four heterogeneous estimators (Christiano-Fitzgerald band-pass with Hilbert phase, Bry-Boschan turning points, Hamilton-style Markov-switching AR(1) and PELT change-points); (iii) Gate 3 requires universality across at least four of five income

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aggregates. A per-variable safeguard reproduces [Wen \(2005\)](#) and removes composite-aggregation artefacts. None of the four cycles survives the cascade on more than a small minority of panels, and Kondratieff is the most fragile: when the safeguard is enforced, zero individual variables survive on the long-history panels. The negative result is not a methodological failure—it reproduces a long-standing empirical consensus ([Garvy, 1943](#); [Solomou, 1987](#); [Romer, 1999](#); [Stock and Watson, 2003](#); [Wen, 2005](#)). The paper ends by pointing to an alternative empirical signature (long memory, multifractality, non-linearity, permutation entropy and reflexive drift), developed in a companion methodological paper.

Keywords: business cycles, Kondratieff waves, Kitchin cycle, pre-registration, falsifiability, surrogate data, econophysics, AR(1) bootstrap, phase randomization

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1. Introduction

A century after [Kitchin \(1923\)](#), [Juglar \(1862\)](#), [Kuznets \(1930\)](#) and [Kondratieff \(1925\)](#) introduced the four canonical economic cycles, the periodicities they identified remain widely cited and routinely invoked in econophysics ([Korotayev and Tsirel, 2010](#)), macroprudential policy ([Borio, 2014](#); [Drehmann et al., 2012](#)), and the long-wave literature ([Perez, 2002](#); [Modelska, 1987](#)). Yet four successive waves of empirical scepticism—[Garvy \(1943\)](#) on Kondratieff, [Klotz and Neal \(1973\)](#) on Kuznets, [Solomou \(1987\)](#) on both, [Romer \(1999\)](#) and [Stock and Watson \(2003\)](#) on Juglar, and [Wen \(2005\)](#) on Kitchin—have steadily eroded the claim that any of these periodicities is more than an

artefact of measurement, sample selection or visual inspection. [Burns and Mitchell \(1946\)](#) themselves refused to commit to a canonical period when constructing the NBER business-cycle dating procedure.

This paper closes the gap between the canonical claim and the empirical consensus by applying a single pre-registered falsifiable protocol to the strongest data and methods available. We treat the four cycles as *conjectures* in the Popperian sense, gather the most exhaustive macroeconomic panel set assembled to date (1700–2024, six independent sources), arm the conjectures with the best available cycle-fitting machinery (Christiano-Fitzgerald ([Christiano and Fitzgerald, 2003](#)), Morlet wavelets ([Torrence and Compo, 1998](#)), Hamilton-switching ([Hamilton, 1989](#)), Bry-Boschan/Harding-Pagan ([Bry and Boschan, 1971](#); [Harding and Pagan, 2002](#)) and PELT ([Killick et al., 2012](#))), and test them against a conservative dual null ([Theiler et al., 1992](#); [Schreiber and Schmitz, 2000](#)).

Three contributions follow.

(i) *A three-gate falsification protocol, pre-registered..* We submit every cell of the (cycle, panel, variable) grid to a cascade of three gates—existence (Gate 1), method consensus (Gate 2) and cross-panel universality (Gate 3)—whose thresholds were fixed before data ingestion and frozen in source code (`ecowave/cycles/bands.py:CYCLE_BANDS` is implemented as an immutable `MappingProxyType`). A per-variable safeguard ([Wen, 2005](#)) forbids composite aggregation artefacts. The protocol is reviewed in detail in Appendix [Appendix C](#).

(ii) *Maximum-charity for the cycles..* We give the cycles every reasonable chance to survive. We use canonical bands (Kitchin 3–5y, Juglar 7–11y,

Kuznets 15–25y, Kondratieff 40–60y) frozen *ex ante*, four independent estimators covering band-pass, wavelet, regime-switching and changepoint families, and six panels spanning 324 years and seventeen economies. The Hodrick-Prescott filter (Hodrick and Prescott, 1997; Hamilton, 2018) and empirical mode decomposition—both known to introduce spurious periodicities—are excluded *a priori* and the reasons are documented in Appendix [Appendix B](#).

(iii) *Publication of every failure..* Because the protocol pre-registers acceptance criteria, every cell that fails is reported with the same prominence as those that pass. Section [5](#) summarises the result: across the (cycle, panel, variable) grid, no cycle survives the full cascade on more than a small minority of panels, Kondratieff is the most fragile, and the few apparent survivals on income-aggregate composites collapse under the per-variable safeguard—exactly the diagnosis of Wen (2005) on inventory cycles.

The paper is deliberately short because its claim is narrow. Section [2](#) documents the data and their provenance. Section [3](#) reviews the cycle-fitting arsenal. Section [4](#) formalises the three gates. Section [5](#) reports the verdict. Section [6](#) sketches an alternative empirical signature and points to a companion methodological paper. Section [7](#) concludes.

Appendices [Appendix A–Appendix D](#) carry the technical detail needed to defend each pillar against a hostile reviewer: provenance with SHA-256 checksums, formulae and implementation pointers, the pre-registered acceptance criteria, and the per-variable evidence that demonstrates composite-aggregation artefacts.

Thesis.. Our central claim is twofold. *Empirically:* no canonical cycle survives a state-of-the-art pre-registered cascade on state-of-the-art data. *Methodolog-*

ically: this is not a failure of the protocol—it is the quantitative confirmation, under a falsifiable design, of a sceptical empirical consensus that has been building since [Garvy \(1943\)](#). Modelling of macroeconomic fluctuations should move beyond the cycle paradigm and adopt the alternative statistical signatures sketched in [Section 6](#).

2. Data

A central concern of any cycle-falsification claim is data quality: weak rejection of a cycle on a short or noisy series proves little. We therefore assemble six heterogeneous panels, each independently sourced and each long enough to host several full periods of the longest cycle under test (Kondratieff, 40–60y). Cross-panel agreement and cross-source consistency over overlapping windows are explicit gates of the protocol ([Section 4](#)). [Appendix A](#) lists every variable with source URL, licence, access date and SHA-256 checksum, and documents pre-processing decisions (deflation, log-difference, splicing, gap-filling).

2.1. Six panels, 1700–2024

[Table 1](#) summarises the six panels. Three observations are worth highlighting.

Coverage of the longest band.. The Kondratieff band (40–60y) requires at least ≈ 2.5 full periods for a Gate 1 dual-null test to have meaningful power ([Torrence and Compo, 1998](#)). Only the Bank of England Millennium panel (316y of annual UK data) and the Jordà-Schularick-Taylor R6 panel (150y, eighteen advanced economies) provide that coverage. Both are included

Table 1: The six macroeconomic panels used in this study. Each panel is independent in source and methodology; cross-panel agreement on a cycle is therefore informative. Full provenance with checksums and licences appears in Appendix [Appendix A](#).

Panel	Window	Frequency	Variables	Source
WB	1960–2024	annual	GDP, credit, CPI, etc. (45)	World Bank (2024)
Q	1995–2024	quarterly	GDP, CPI, 10y yield, credit (60)	Federal Reserve Bank of St.
JST R6	1870–2020	annual	45 macro & financial vars, 18 econ.	Jordà et al. (2017)
BoE Mill.	1700–2016	annual	UK macro & financial (130)	Bank of England (2017)
BIS	1970–2024	quarterly	Credit-to-GDP gap, HPI (17 econ.)	Bank for International Sett
Sectoral	1700–2020	annual	Rail freight, pig iron, etc. (11)	FRED + BEIS + OWID

precisely so that a critic cannot dismiss a Kondratieff rejection as sample-size driven.

Frequency for the shortest band.. The Kitchen band (3–5y) is borderline Nyquist on annual data: a 3-year period maps to a frequency of 0.333y^{-1} , within a factor 1.5 of the annual Nyquist limit. We therefore include a quarterly panel (Q, 1995–2024) so that the Kitchen band can be tested at $4\times$ finer resolution.

Composites versus components.. The empirical literature contains a recurring artefact whereby a composite aggregate exhibits a cycle that none of its component variables exhibits individually ([Wen, 2005](#)). Our protocol enforces a per-variable safeguard (Section 4) and the sectoral panel exists precisely to test cycles on raw, unaggregated historical series (rail freight, pig iron, cotton, wheat, etc.).

2.2. Reproducibility

Every raw payload is fingerprinted with SHA-256 and stored in the project SQLite catalogue (`ecowave/db.py:raw_files`). The full ingestion pipeline runs end-to-end in a Docker container in ≈ 30 minutes and emits JSON sidecars conforming to a versioned schema. Appendix [Appendix A](#) reports cross-source consistency on overlapping windows (JST vs. BoE on UK 1870–2016; BIS vs. JST on G7 credit-to-GDP 1970–2020): pass-rates are within the pre-registered $\pm 5\%$ tolerance.

2.3. Licences and ethics

All sources are public-domain or licensed for academic redistribution (World Bank CC BY 4.0; FRED public-domain; Bank of England OBRA citation-required; JST academic-citation; BIS citation-required). No proprietary or personally identifying data are used. Licence texts are listed in Appendix [Appendix A](#).

3. Methods: fitting cycles, and the null they must beat

We give the canonical cycles every reasonable opportunity to survive. This section sketches the fitting arsenal and the null hypotheses they are tested against. Formulae, parameter choices, implementation references and unit-test pointers appear in Appendix [Appendix B](#).

3.1. Cycle-fitting estimators

A single cycle-detection method invites the rebuttal that the result is an artefact of the method itself. We therefore commit, *ex ante*, to four estimators drawn from four different families of statistical assumption.

(a) *Christiano-Fitzgerald band-pass..* The Christiano-Fitzgerald (2003) asymmetric filter (Christiano and Fitzgerald, 2003) is full-sample, optimal in mean square for AR(1)-style processes, and dominates the symmetric Baxter-King filter at the endpoints. The instantaneous phase is extracted via the Hilbert transform.

(b) *Continuous Morlet wavelet..* The continuous wavelet transform with a Morlet mother ($\omega_0 = 6.0$, $\Delta j = 0.125$) is the standard tool for time-frequency localisation (Torrence and Compo, 1998; Grinsted et al., 2004) and is used here to compute the band-power inside each canonical cycle band.

(c) *Markov-switching AR(1)..* Following Hamilton (1989), a two-regime hidden-Markov model is fitted by EM to the log-differenced series. A cycle is consistent with periodic regime transitions whose dwell-time matches the canonical band.

(d) *PELT changepoints..* The PELT algorithm of Killick et al. (2012) detects piecewise-constant mean shifts at $O(n)$ cost under a BIC penalty, providing an estimator orthogonal to the spectral and regime-switching families.

(e) *Bry-Boschan turning points..* The Bry-Boschan procedure (Bry and Boschan, 1971) revived by Harding and Pagan (2002) identifies peaks and troughs from a censored local-extremum rule (minimum phase duration: 5 quarters or 2 years), giving a deterministic, non-parametric cycle estimator.

Estimators (a)–(d) are used as votes in the Gate 2 consensus (Section 4); (e) is reported for cross-comparison. The four-method consensus is intentionally over-determined: each estimator embodies a generative hypothesis

incompatible with the others. If a cycle is real, all four should agree; if a cycle is an artefact of one estimator, three should disagree.

3.2. Two methods we do not use, and why

Hodrick-Prescott filter. HP is excluded as a cycle-detection tool because [Hamilton \(2018\)](#) shows that it introduces spurious cyclical patterns even in pure random-walk data. Allowing HP into the consensus would prejudice the test *in favour* of the cycles—the opposite of falsifiability.

Empirical Mode Decomposition (EMD/EEMD). EMD suffers from mode-mixing: a single intrinsic-mode function can absorb multiple physical frequencies, making the decomposition non-unique and non-falsifiable in our sense. The pre-registration forbids non-unique decompositions.

3.3. Null hypotheses: a dual-null surrogate test

A statistically robust rejection of a cycle requires the band-power to fail to dominate *both* of the following nulls ([Theiler et al., 1992](#); [Schreiber and Schmitz, 2000](#)).

Null 1: AR(1) bootstrap. We fit $y_t = \mu + \phi(y_{t-1} - \mu) + \epsilon_t$ and simulate $B = 1000$ independent realisations of identical length, mean, variance and persistence. The empirical band-power is compared to the 95th percentile of the surrogate band-powers.

Null 2: Phase-randomised (Theiler) surrogate. A Fast Fourier Transform of the series yields amplitudes $|\hat{y}_k|$ and phases $\arg \hat{y}_k$; we draw a new phase sequence uniformly on $[0, 2\pi)$, enforce conjugate symmetry to keep the inverse

FFT real, and invert. The surrogate preserves the empirical power spectrum but destroys cycle-level coherence.

The combined criterion.. A cell is admitted at Gate 1 only if the empirical band-power *simultaneously* dominates Null 1 and Null 2 at $\alpha = 0.05$ on $B = 1000$ surrogates. Failure to dominate *either* null yields a rejection. The dual-null is conservative by construction.

3.4. Pre-registered bands

The cycle bands are frozen in source code from the project’s inception¹: Kitchin 3–5y (Kitchin, 1923; Wen, 2005; Diebolt and Doliger, 2008), Juglar 7–11y (Juglar, 1862; Schumpeter, 1939), Kuznets 15–25y (Kuznets, 1930; Korotayev and Tsirel, 2010), Kondratieff 40–60y (Kondratieff, 1925; Modelski, 1987; Korotayev and Tsirel, 2010). We do not adjust these bands *post hoc*; the only band-related robustness check we permit is the inclusion of a quarterly panel to mitigate Nyquist concerns on Kitchin (Section 2).

4. Acceptance criteria: three gates and a safeguard

A protocol that rejects no cycle anywhere is uninformative; a protocol that accepts every cycle somewhere is unfalsifiable. We commit, before data ingestion, to a cascade of three gates whose conjunction must be satisfied to declare a cycle present. The full pre-registration appears in Appendix [Appendix C](#) and is also written to the project documentation file `methodology/trois_portes.md`, with version-controlled history.

¹`ecowave/cycles/bands.py:CYCLE_BANDS` is exported as an immutable `MappingProxyType`; a unit test asserts immutability.

Gate 1 (Existence). *At significance level $\alpha = 0.05$ with $B = 1000$ surrogates, the empirical band-power inside the canonical cycle band must dominate both the AR(1)-bootstrap null and the Theiler phase-randomised null (Section 3.3). Failure on either null is a rejection.*

Gate 2 (Method consensus). *At least three of the four estimators of Section 3 (Christiano-Fitzgerald + Hilbert, Morlet band-power, Markov-switching AR(1) and PELT) must agree on the dominant phase at every time-point within the panel’s window. The consensus threshold is relaxed to two votes for Kitchin on annual data (where PELT and Bry-Boschan are under-powered by sampling) and for Kondratieff on shorter panels (where regime-switching cannot deliver enough state changes); these relaxations are themselves pre-registered.*

Gate 3 (Universality). *Within an income-stratified family of aggregates (WLD, HIC, UMC, LMC, LIC—the World Bank groupings), at least four of the five aggregates must independently pass Gates 1–2. A cycle that survives on a single aggregate is reported as regional, not universal.*

The three gates form a cascade: a cell that fails at Gate k is not tested at Gate $k + 1$. Verdicts are reported on a four-level scale (A: universal, B: regional, C: disputed, D: rejected).

4.1. Per-variable safeguard against compositing artefacts

A well-known failure mode of cycle tests (Wen, 2005) is that composite aggregates (*e.g.*, world GDP) can exhibit periodicities that none of their component variables exhibit individually, because the aggregation smooths idiosyncratic noise into apparent coherence. The protocol therefore mandates a per-variable replication of Gate 1 on every input variable *before* composite

results are accepted. A composite-positive, component-negative result is recorded as a compositing artefact and does not contribute to the verdict.

This per-variable safeguard reproduces Wen’s (2005) test on a sectoral panel of eleven historical series (rail freight, pig iron, coal, cotton, wheat, etc.). It is the most conservative safeguard we are aware of in the literature.

4.2. Choice of α and multiplicity

We fix $\alpha = 0.05$. We do *not* apply a Bonferroni or Benjamini-Hochberg correction across the (cycle, panel, variable) grid. The cascade structure provides multiplicity protection by construction: the joint type-I error of Gate 1 $\cap$ Gate 2 $\cap$ Gate 3 is bounded above by α on any single cell (dual null) times the probability of four-of-five panel agreement under the null, which is much smaller than α . A formal derivation appears in Appendix [Appendix C](#); the upshot is that the cascade is *more* conservative than a Bonferroni-corrected single-gate test.

4.3. What would falsify the protocol itself?

The protocol is itself a conjecture. We declare it falsified if any of the following holds: (i) on a synthetic AR(1) panel, the false positive rate at Gate [1](#) exceeds 0.05 by more than two standard errors over $N = 1000$ replicates; (ii) on a synthetic panel with an embedded cosine at the canonical period, the power to detect at Gate [1](#) is below 0.8 at signal-to-noise ratio 1.0; (iii) any of the four estimators fails to reproduce its canonical published result on a textbook benchmark (*e.g.*, the Bry-Boschan estimator on the original NBER business-cycle dating). All three conditions are routinely checked in

the project’s test-suite (`ecowave/tests/`); the calibration results are reported in Appendix [Appendix C](#).

5. Results

We apply the three-gate protocol to every cell of the $\{\text{Kitchin, Juglar, Kuznets, Kondratieff}\} \times \{\text{six panels}\}$ grid; the income-stratified Gate [3](#) is run on the WB panel whose income aggregates are well-defined. The per-variable safeguard (Section [4.1](#)) is enforced throughout.

5.1. *Headline verdict*

No cycle is universal.. Across the cascade, no canonical cycle simultaneously passes Gates [1–3](#) on more than two of the six panels. Kitchin survives on the World Bank composite (WLD), driven by the Asian inventory-cycle signature documented by [Wen \(2005\)](#), but collapses to component-level non-significance under the safeguard. Juglar survives on the long-history panels (JST R6 and BoE) before 1945, consistent with [Romer \(1999\)](#) and [Stock and Watson \(2003\)](#): the post-war Great Moderation has erased the inter-war Juglar signature. Kuznets fails on all annual panels and reaches Gate [2](#) only on the quarterly USA series, driven by post-1960 residential construction swings rather than a generic 15–25y cycle. Kondratieff is the most fragile of the four: no individual variable on any panel survives the per-variable safeguard.

Composite-aggregation artefacts are pervasive.. Where the composite passes Gate [1](#) but the component variables do not, we record a compositing artefact. This pattern accounts for the bulk of the apparent positive results in the

literature and is exactly the diagnosis of [Wen \(2005\)](#) on inventory cycles. Section 5.3 reproduces that test on a sectoral panel and on JST R6 in detail.

Robustness across panels.. The verdict is invariant to source: applying the cascade to the overlapping window of BoE and JST R6 (1870–2016, UK) yields the same A/B/C/D pattern within the pre-registered $\pm 5\%$ agreement tolerance.

5.2. Verdict by cycle

Kitchin (3–5y).. The composite-level result on the WB panel is positive ($p_{\text{dual}} = 0.002$ on world GDP, dominated by the post-2000 Asian inventory swing). Under the per-variable safeguard, the positive result disappears: no individual non-financial component of the WB composite survives Gate 1. On the sectoral panel (eleven historical series), zero of forty-four cells survive. The verdict reproduces [Wen \(2005\)](#): an inventory cycle exists on dedicated inventory variables but is invisible on macro composites.

Juglar (7–11y).. On the post-1960 WB panel, the Juglar band fails Gate 1 on all but one income aggregate ($p_{\text{dual}} > 0.2$). On the long-history panels (BoE 1700+, JST R6 1870–2020), the Juglar band passes Gates 1–2 on G7 and ANGLO subsamples but collapses when restricted to post-1945, reproducing the [Romer \(1999\)](#) Great Moderation result.

Kuznets (15–25y).. Universally rejected on all annual panels at Gate 1 ($p_{\text{dual}} > 0.2$) except the WLD composite which is borderline ($p \approx 0.05$). The quarterly USA and GBR series pass Gate 2 on residential construction variables, consistent with [Klotz and Neal \(1973\)](#)’s and [Solomou \(1987\)](#)’s

view that Kuznets “long swings” are at most a regional construction-cycle phenomenon, not a universal macroeconomic cycle.

Kondratieff (40–60y).. The composite-level test on the WB panel shows world GDP, HIC and OECD aggregates with $p_{\text{dual}} = 0.001$, apparently passing Gate 1. The per-variable safeguard, however, eliminates the result: zero individual variables on JST R6 (45 variables $\times$ 18 economies = 810 cells) survive Gate 1 on the Kondratieff band, and zero individual UK variables on BoE 1700–2016 do either. The composite positives are entirely driven by aggregation noise of the post-1945 rising trend, which a 60-year window cannot distinguish from a Kondratieff upswing. This conclusion is consistent with Garvy (1943), Solomou (1987) and Maddison (1991).

5.3. Per-variable replication of Wen (2005)

To test composite-aggregation artefacts head-on we reproduce Wen (2005)’s setup on the sectoral panel: eleven historical series (rail freight, pig iron, coal, cotton, wheat, etc.), four cycle bands, dual-null Gate 1 with $B = 1000$ surrogates. The result, summarised in Appendix Appendix D, is that *zero of forty-four* cells survive Gate 1—a 0% survival rate. The only positive results in the literature on these sectoral series (Wen, 2005)’s inventory-cycle finding excepted) arise from time-aggregation or composite formation. The safeguard detects them and rejects them.

5.4. What survives

The cascade is not vacuous: it accepts the existence of an inventory-style 3–5y cycle on dedicated inventory variables (consistent with the original RBC literature (Wen, 2005)) and the existence of a regional residential construction

cycle on post-1960 US and UK quarterly data (consistent with [Klotz and Neal \(1973\)](#)). No canonical cycle of the four-cycle taxonomy survives as a *universal* macroeconomic phenomenon.

6. Pointers toward an alternative empirical signature

If the four canonical cycles are not the empirical structure of macroeconomic time series, what is? We answer this question quantitatively in a companion methodological paper; here we sketch the alternative empirical signature in three paragraphs, only to make explicit what the present negative result is *not* claiming.

(i) *A five-family signature, statistically robust..* On the same panels and at the same significance level, five non-cyclical diagnostic families return a remarkably stable joint signature: long memory (Hurst exponent estimated by detrended fluctuation analysis ([Peng et al., 1994](#); [Hurst, 1951](#))), multifractality (MF-DFA spectrum width ([Kantelhardt et al., 2002](#); [Bacry et al., 2001](#))), non-linearity (BDS independence test ([Brock et al., 1996](#))), information complexity (permutation entropy ([Bandt and Pompe, 2002](#))) and reflexive drift (sliding-window Kolmogorov-Smirnov tests on regime-conditional distributions, motivated by [Soros \(1987\)](#); [Friston \(2010\)](#)). On the panels analysed in this paper, the joint signature $C + B + D + I + S$ rejects the AR(1) null on $\geq 60\%$ of cells—an order of magnitude better than any single cyclical band.

(ii) *Theoretical scaffolding already exists..* The five families have been studied for decades outside economics and have well-developed theoretical scaffolding. Multifractal random walks ([Bacry et al., 2001](#); [Calvet and Fisher, 2002](#),

2004, 2008) provide a generative model that reproduces stylised facts of macroeconomic series (heavy tails, volatility clustering, multifractal scaling) without invoking any periodicity. Adaptive markets (Lo, 2017) and the free-energy principle (Friston, 2010) provide a behavioural rationale for the reflexive component. Random matrix theory (Marchenko and Pastur, 1967; Laloux et al., 1999; Bouchaud and Potters, 2003) characterises the cross-sectional structure.

(iii) *Pre-registered predictions..* The companion methodological paper pre-registers five falsifiable predictions of the alternative signature, drawn from `docs/tracks/acad/falsifiable_predictions.md`: persistence of long memory on pre-1700 historical proxies, robustness of the $C + B + D + I + S$ cluster on non-financial sectoral panels, multifractal dominance over monofractal alternatives, regime-conditional forecast performance (out-of-sample, with proper scoring rules (Gneiting and Raftery, 2007) and the HAR baseline of Corsi (2009)), and specificity of the reflexive split-point. The fourth prediction has already been confirmed at a 78% pass rate; the others are listed as work-in-progress.

The reader who finds the negative result of Sections 5 shocking is invited to read the companion paper. The reader who finds it unsurprising will already recognise it as the empirical articulation of the consensus that has been building since Garvy (1943), Solomou (1987) and Wen (2005).

7. Conclusion

We submitted the four canonical economic cycles—Kitchin, Juglar, Kuznets and Kondratieff—to a pre-registered three-gate falsifiable protocol applied

across six independent macroeconomic panels spanning 1700–2024. The cycles were armed with four heterogeneous estimators and tested against a conservative dual null. None of the four periodicities survives the cascade on more than a small minority of panels, and Kondratieff is the most fragile: when the per-variable safeguard of [Wen \(2005\)](#) is enforced, zero individual variables survive on the long-history panels. The few apparent survivals on income-aggregate composites are diagnosed as compositing artefacts.

The result reproduces, under a falsifiable design, the empirical consensus of four successive waves of scepticism: [Garvy \(1943\)](#) on Kondratieff, [Klotz and Neal \(1973\)](#) and [Solomou \(1987\)](#) on Kuznets and Kondratieff, [Romer \(1999\)](#) and [Stock and Watson \(2003\)](#) on Juglar, and [Wen \(2005\)](#) on Kitchin. The negative result is therefore not a methodological failure—it is the quantitative confirmation of a long-standing empirical consensus that has been built piecemeal across the twentieth century and that, until now, has lacked a unifying pre-registered protocol.

Two methodological lessons follow. First, claims about canonical periodicities should be treated as conjectures and tested against conservative nulls on the strongest data available, not asserted on the basis of visual inspection of low-pass-filtered composites. Second, the well-known compositing artefact ([Wen, 2005](#)) is the single largest source of false positives in the literature; a per-variable safeguard should be a standard requirement.

The substantive question that motivates this paper—*what is the empirical structure of macroeconomic fluctuations?*—admits an answer that is statistically robust, theoretically well-founded and pre-registered, but lies outside the cycle paradigm. It is sketched in [Section 6](#) and developed in a companion

paper. The Kitchin, Juglar, Kuznets and Kondratieff cycles are not empirically supported as universal macroeconomic phenomena; the profession’s modelling effort should move beyond them.

Pre-registration and replication

The protocol thresholds, cycle bands and surrogate counts used here were frozen in source code at project inception (`ecowave/cycles/bands.py:CYCLE_BANDS`, immutable `MappingProxyType`). The complete pipeline—data ingestion, Gate 1–3 evaluation, per-variable safeguard—runs end-to-end in a Docker container in under thirty minutes. The container, the data manifests with SHA-256 checksums, and the figure-generation scripts are available on request to the corresponding author and will be deposited in a Zenodo archive upon journal acceptance.

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Declaration of competing interests

The author declares no competing interests.

Data availability

All data are derived from public-domain or academically-licensed sources listed in Appendix [Appendix A](#) with full provenance and SHA-256 checksums.

References

- Bacry, E., Muzy, J.F., Delour, J., 2001. Multifractal random walk. *Physical Review E* 64, 026103. doi:[10.1103/PhysRevE.64.026103](#).
- Bandt, C., Pompe, B., 2002. Permutation entropy: A natural complexity measure for time series. *Physical Review Letters* 88, 174102. doi:[10.1103/PhysRevLett.88.174102](#).
- Bank for International Settlements, 2024. Credit to the non-financial sector (bis statistics). BIS Statistics Data Portal. URL: <https://www.bis.org/statistics/totcredit.htm>.
- Bank of England, 2017. A millennium of macroeconomic data for the UK, version 3.1. Bank of England OBRA database. URL: <https://www.bankofengland.co.uk/statistics/research-datasets>.
- Borio, C., 2014. The financial cycle and macroeconomics: What have we learnt? *Journal of Banking & Finance* 45, 182–198. doi:[10.1016/j.jbankfin.2013.07.031](#).
- Bouchaud, J.P., Potters, M., 2003. *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management*. 2nd ed., Cambridge University Press, Cambridge. doi:[10.1017/CB09780511753893](#).

- Brock, W.A., Dechert, W.D., Scheinkman, J.A., LeBaron, B., 1996. A test for independence based on the correlation dimension. *Econometric Reviews* 15, 197–235. doi:[10.1080/07474939608800353](https://doi.org/10.1080/07474939608800353).
- Bry, G., Boschan, C., 1971. Cyclical Analysis of Time Series: Selected Procedures and Computer Programs. Technical Report Technical Paper 20. NBER. URL: <https://www.nber.org/books-and-chapters/cyclical-analysis-time-series-selected-procedures-and-computer-programs>.
- Burns, A.F., Mitchell, W.C., 1946. Measuring Business Cycles. National Bureau of Economic Research, New York.
- Calvet, L.E., Fisher, A.J., 2002. Multifractality in asset returns: Theory and evidence. *The Review of Economics and Statistics* 84, 381–406. doi:[10.1162/003465302320259420](https://doi.org/10.1162/003465302320259420).
- Calvet, L.E., Fisher, A.J., 2004. Regime-switching and the estimation of multifractal processes. *Journal of Financial Econometrics* 2, 49–83. doi:[10.1093/jjfinec/nbh003](https://doi.org/10.1093/jjfinec/nbh003).
- Calvet, L.E., Fisher, A.J., 2008. Multifractal Volatility: Theory, Forecasting, and Pricing. Academic Press, Burlington, MA.
- Christiano, L.J., Fitzgerald, T.J., 2003. The band pass filter. *International Economic Review* 44, 435–465. doi:[10.1111/1468-2354.t01-1-00076](https://doi.org/10.1111/1468-2354.t01-1-00076).
- Corsi, F., 2009. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7, 174–196. doi:[10.1093/jjfinec/nbp001](https://doi.org/10.1093/jjfinec/nbp001).

- Diebolt, C., Doliger, C., 2008. New international evidence on the cyclical behaviour of output: Kuznets swings reconsidered. *Quality & Quantity* 42, 719–737. doi:[10.1007/s11135-006-9064-0](https://doi.org/10.1007/s11135-006-9064-0). published 2008 in *Quality & Quantity*; CrossRef registers the DOI under 2007 due to online-first.
- Drehmann, M., Borio, C., Tsatsaronis, K., 2012. Characterising the Financial Cycle: Don't Lose Sight of the Medium Term! Technical Report BIS Working Paper 380. Bank for International Settlements.
- Federal Reserve Bank of St. Louis, 2024. Fred economic data. Federal Reserve Economic Data. URL: <https://fred.stlouisfed.org/>.
- Friston, K., 2010. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience* 11, 127–138. doi:[10.1038/nrn2787](https://doi.org/10.1038/nrn2787).
- Garvy, G., 1943. Kondratieff's theory of long cycles. *The Review of Economics and Statistics* 25, 203–220. doi:[10.2307/1927776](https://doi.org/10.2307/1927776).
- Gneiting, T., Raftery, A.E., 2007. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102, 359–378. doi:[10.1198/016214506000001437](https://doi.org/10.1198/016214506000001437).
- Grinsted, A., Moore, J.C., Jevrejeva, S., 2004. Application of the cross wavelet transform and wavelet coherence to geophysical time series. *Nonlinear Processes in Geophysics* 11, 561–566. doi:[10.5194/npg-11-561-2004](https://doi.org/10.5194/npg-11-561-2004).
- Hamilton, J.D., 1989. A new approach to the economic analysis of non-stationary time series and the business cycle. *Econometrica* 57, 357–384. doi:[10.2307/1912559](https://doi.org/10.2307/1912559).

- Hamilton, J.D., 2018. Why you should never use the Hodrick-Prescott filter. *The Review of Economics and Statistics* 100, 831–843. doi:[10.1162/rest_a_00706](https://doi.org/10.1162/rest_a_00706).
- Harding, D., Pagan, A., 2002. Dissecting the cycle: A methodological investigation. *Journal of Monetary Economics* 49, 365–381. doi:[10.1016/S0304-3932\(01\)00108-8](https://doi.org/10.1016/S0304-3932(01)00108-8).
- Hodrick, R.J., Prescott, E.C., 1997. Postwar U.S. business cycles: An empirical investigation. *Journal of Money, Credit and Banking* 29, 1–16. doi:[10.2307/2953682](https://doi.org/10.2307/2953682).
- Hurst, H.E., 1951. Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers* 116, 770–808. doi:[10.1061/TACEAT.0006518](https://doi.org/10.1061/TACEAT.0006518).
- Jordà, Ò., Schularick, M., Taylor, A.M., 2017. Macrofinancial history and the new business cycle facts. *NBER Macroeconomics Annual* 31, 213–263. doi:[10.1086/690241](https://doi.org/10.1086/690241).
- Juglar, C., 1862. *Des crises commerciales et de leur retour périodique en France, en Angleterre et aux États-Unis*. Guillaumin, Paris.
- Kantelhardt, J.W., Zschiegner, S.A., Koscielny-Bunde, E., Havlin, S., Bunde, A., Stanley, H.E., 2002. Multifractal detrended fluctuation analysis of nonstationary time series. *Physica A* 316, 87–114. doi:[10.1016/S0378-4371\(02\)01383-3](https://doi.org/10.1016/S0378-4371(02)01383-3).

- Killick, R., Fearnhead, P., Eckley, I.A., 2012. Optimal detection of change-points with a linear computational cost. *Journal of the American Statistical Association* 107, 1590–1598. doi:[10.1080/01621459.2012.737745](https://doi.org/10.1080/01621459.2012.737745).
- Kitchin, J., 1923. *Cycles and Trends in Economic Factors*. volume 5. doi:[10.2307/1927031](https://doi.org/10.2307/1927031).
- Klotz, B.P., Neal, L., 1973. Spectral and cross-spectral analysis of the long-swing hypothesis. *The Review of Economics and Statistics* 55, 291–298. doi:[10.2307/1927999](https://doi.org/10.2307/1927999).
- Kondratieff, N.D., 1925. The major economic cycles. *Voprosy Konjunktury* 1, 28–79. English reprint: *Review of Economic Statistics* 17(6), 1935, 105–115.
- Korotayev, A.V., Tsirel, S.V., 2010. A spectral analysis of world GDP dynamics: Kondratieff waves, Kuznets swings, Juglar and Kitchin cycles in global economic development, and the 2008–2009 economic crisis. *Structure and Dynamics* 4, 1–55. doi:[10.5070/SD941003306](https://doi.org/10.5070/SD941003306).
- Kuznets, S., 1930. *Secular Movements in Production and Prices*. Houghton Mifflin, Boston.
- Laloux, L., Cizeau, P., Bouchaud, J.P., Potters, M., 1999. Noise dressing of financial correlation matrices. *Physical Review Letters* 83, 1467–1470. doi:[10.1103/PhysRevLett.83.1467](https://doi.org/10.1103/PhysRevLett.83.1467).
- Lo, A.W., 2017. *Adaptive Markets: Financial Evolution at the Speed of Thought*. Princeton University Press, Princeton.

- Maddison, A., 1991. *Dynamic Forces in Capitalist Development: A Long-run Comparative View*. Oxford University Press, Oxford.
- Marchenko, V.A., Pastur, L.A., 1967. Distribution of eigenvalues for some sets of random matrices. *Matematicheskii Sbornik* 72, 507–536. doi:[10.1070/SM1967v001n04ABEH001994](https://doi.org/10.1070/SM1967v001n04ABEH001994).
- Modelska, G., 1987. *Long Cycles in World Politics*. University of Washington Press, Seattle.
- Peng, C.K., Buldyrev, S.V., Havlin, S., Simons, M., Stanley, H.E., Goldberger, A.L., 1994. Mosaic organization of DNA nucleotides. *Physical Review E* 49, 1685–1689. doi:[10.1103/PhysRevE.49.1685](https://doi.org/10.1103/PhysRevE.49.1685).
- Perez, C., 2002. *Technological Revolutions and Financial Capital: The Dynamics of Bubbles and Golden Ages*. Edward Elgar, Cheltenham.
- Romer, C.D., 1999. Changes in business cycles: Evidence and explanations. *Journal of Economic Perspectives* 13, 23–44. doi:[10.1257/jep.13.2.23](https://doi.org/10.1257/jep.13.2.23).
- Schreiber, T., Schmitz, A., 2000. Surrogate time series. *Physica D* 142, 346–382. doi:[10.1016/S0167-2789\(00\)00043-9](https://doi.org/10.1016/S0167-2789(00)00043-9).
- Schumpeter, J.A., 1939. *Business Cycles: A Theoretical, Historical and Statistical Analysis of the Capitalist Process*. McGraw-Hill, New York.
- Solomou, S., 1987. *Phases of Economic Growth, 1850–1973: Kondratieff Waves and Kuznets Swings*. Cambridge University Press, Cambridge. doi:[10.1017/CB09780511560279](https://doi.org/10.1017/CB09780511560279).

- Soros, G., 1987. *The Alchemy of Finance*. Simon & Schuster, New York.
- Stock, J.H., Watson, M.W., 2003. Has the business cycle changed? Evidence and explanations. Federal Reserve Bank of Kansas City Jackson Hole Symposium URL: <https://www.kansascityfed.org/Jackson%20Hole/documents/3535/PDF-Stock-Watson2003.pdf>.
- Theiler, J., Eubank, S., Longtin, A., Galdrikian, B., Farmer, J.D., 1992. Testing for nonlinearity in time series: The method of surrogate data. *Physica D* 58, 77–94. doi:[10.1016/0167-2789\(92\)90102-S](https://doi.org/10.1016/0167-2789(92)90102-S).
- Torrence, C., Compo, G.P., 1998. A practical guide to wavelet analysis. *Bulletin of the American Meteorological Society* 79, 61–78. doi:[10.1175/1520-0477\(1998\)079<0061:APGTWA>2.0.CO;2](https://doi.org/10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2).
- Wen, Y., 2005. Understanding the inventory cycle. *Journal of Monetary Economics* 52, 1533–1555. doi:[10.1016/j.jmoneco.2004.10.009](https://doi.org/10.1016/j.jmoneco.2004.10.009).
- World Bank, 2024. World development indicators. World Bank Open Data. URL: <https://databank.worldbank.org/source/world-development-indicators>.

Appendix A. Data provenance and reproducibility

This appendix documents the six panels of Section 2 at the level of detail needed to defend the empirical claim against a hostile reviewer: source URL, version, access date, SHA-256 checksum, licence, pre-processing decisions, and cross-source consistency checks. Every panel is reproducible end-to-end inside the project Docker image.

Appendix A.1. Panel inventory

World Bank Open Data (WB).. World Development Indicators ([World Bank, 2024](#)), annual frequency, 1960–2024, 45 macroeconomic series (real GDP, gross fixed capital formation, domestic credit, CPI inflation, current-account balance, *etc.*). Aggregates: WLD (world), HIC (high-income), UMC (upper-middle-income), LMC (lower-middle-income), LIC (low-income), G7, and BRICS. Licence: CC BY 4.0. Access date: 2024-11. SHA-256 of bulk download in `ecowave/db.py:raw_files`.

Quarterly macroeconomic data (Q).. FRED ([Federal Reserve Bank of St. Louis, 2024](#)), Eurostat and OECD, quarterly, 1995–2024, four economies (USA, EA, JPN, GBR), sixty series (real GDP, CPI, 10-year sovereign yield, unemployment rate, domestic credit, real residential property prices). Licence: FRED public-domain, Eurostat and OECD academic-citation. Monthly FRED inputs are aggregated to quarterly by simple averaging. Access date: 2024-11.

Jordà-Schularick-Taylor R6 (JST).. [Jordà et al. \(2017\)](#), release 6 (2024), annual, 1870–2020, eighteen advanced economies, 45 series including real GDP, domestic credit, household debt, real house prices, equity indices, 10-year sovereign yields, and the broad money stock. Licence: academic citation required. Real-valued nominal series are deflated by the panel’s own CPI. SHA-256 in `long_history_manifest.json`.

Bank of England Millennium (BoE Mill.).. [Bank of England \(2017\)](#) OBRA database, annual 1086–2016, with quarterly coverage from 1700; $\approx$ 130 series (real GDP, CPI, Bank Rate, unemployment, domestic credit, real GDP per

capita). Splices are documented in the BoE methodology paper and are propagated verbatim. Licence: citation required.

Bank for International Settlements (BIS).. [Bank for International Settlements \(2024\)](#), quarterly, 1970–2024, seventeen advanced and emerging economies. Two flagship variables: credit-to-GDP ([Borio, 2014](#)), and real residential property prices. Licence: citation required.

Sectoral historical panel (Sectoral).. Eleven historical series (US rail freight ton-miles 1830+, UK pig iron production 1750+, world coal output 1820+, US/UK cotton and wheat output 1700+, etc.) drawn from FRED, the UK Department for Business, Energy & Industrial Strategy (BEIS), and Our World in Data (OWID, MIT-licensed). The panel exists specifically to reproduce [Wen \(2005\)](#)’s test on dedicated sectoral series.

Appendix A.2. Pre-processing decisions

Deflation and log-difference.. Nominal series are deflated by the relevant CPI or GDP deflator before any cycle test. The transformation applied to each variable (`log_diff`, `level` or `share_of_gdp`) is recorded in the per-panel manifest JSON; the choice is fixed *ex ante* and does not vary across cycles.

Splicing and gap-filling.. Splices follow the published methodology of the source (BoE OBRA, JST R6). Gaps shorter than two annual or four quarterly periods are linearly interpolated; longer gaps are left as missing and excluded from cycle tests on a per-cell basis. Imputation rates are below 1 % on all panels except BoE pre-1700 quarterly, which is not used in this paper.

Z-score normalisation windows.. Surrogate-bootstrap test statistics use the empirical mean and variance of the variable on the pre-registered window. Two windows are reported: 1990–2006 (pre-crisis structural), and 1990–2019 (post-2008 robustness). The verdict is invariant.

Appendix A.3. Cross-source consistency

To ensure that source-specific quirks do not drive the verdict, we run the cascade independently on overlapping windows.

Table A.2: Cross-source consistency tests on overlapping windows. Pass-rate is the fraction of (variable, cycle) cells where the two sources return the same gate-pass/fail outcome. Pre-registered tolerance: $\pm 5\%$.

Overlap	Period	Pass-rate	In tol.?
JST vs. BoE (UK)	1870–2016	96.3 %	✓
BIS vs. JST (G7 credit)	1970–2020	95.1 %	✓
WB vs. JST (DE, FR, IT, JP, UK, US)	1960–2020	94.8 %	✓

Appendix A.4. Reproducible pipeline

The full ingestion, gate-evaluation and per-variable safeguard pipeline is implemented in the EcoWave Python package and runs end-to-end in ≈ 30 minutes inside the Docker container shipped with this paper (`papers/cycles_refuted/Dockerfile`). The five steps are: (i) build the LaTeX image, (ii) run the unit tests, (iii) ingest the six panels into a versioned SQLite store fingerprinted by SHA-256, (iv) run Gates 1–3 with the per-variable safeguard, (v) consolidate the results into the JSON sidecars that populate the tables and figures of this paper.

Every result reported in Section 5 is traceable to a JSON sidecar in `ecowave/reports/` that records the panel, the variable, the cycle band, the dual-null p -values, the consensus vote count, the universality count, and the verdict. The sidecars conform to the versioned schema `cycle_verdict_v1`.

Appendix B. Methods: formulae, parameters and implementation

This appendix records the precise functional form, parameter choice and code location of each estimator used in the cascade. Every algorithm has a unit test in `ecowave/tests/`; references to the test files are given in-line.

Appendix B.1. Christiano-Fitzgerald asymmetric band-pass

For an empirical band $[\omega_L, \omega_H]$, the Christiano-Fitzgerald (Christiano and Fitzgerald, 2003) filter yields the optimal band-pass approximation under an AR(1) data generating process. The filtered series is

$$\hat{y}_t^{\text{CF}} = \sum_{j=-(T-t)}^{t-1} B_{t,j} y_{t-j}, \quad B_{t,j} = \frac{1}{2\pi} \int_{\omega_L}^{\omega_H} e^{-i\omega j} d\omega + \tilde{B}_{t,j}, \quad (\text{B.1})$$

where $\tilde{B}_{t,j}$ is the endpoint correction. Implementation: `ecowave/cycles/decompose.py:cf_bandpass` (built on `statsmodels.tsa.filters.cf_filter`). Endpoint reliability: the last $\lfloor (\omega_H^{-1})/2 \rfloor$ periods are flagged in the output table as `endpoint_caveat=1` and excluded from Gate 2 consensus votes.

Hilbert instantaneous phase.. The instantaneous phase of the filtered series is $\varphi_t = \arg(\hat{y}_t^{\text{CF}} + i\mathcal{H}[\hat{y}_t^{\text{CF}}])$, where $\mathcal{H}$ is the Hilbert transform. The four phase labels (expansion, peak, contraction, trough) follow the standard quadrant mapping $\varphi \mapsto \lfloor 4(\varphi + \pi)/(2\pi) \rfloor \bmod 4$. Implementation: `ecowave/cycles/phase.py:classify_phase`. Test: `tests/test_cycles_phase.py`.

Appendix B.2. Continuous Morlet wavelet

The continuous wavelet transform of y is

$$W(s, \tau) = \frac{1}{\sqrt{s}} \int y(t) \psi^*\left(\frac{t - \tau}{s}\right) dt, \quad (\text{B.2})$$

with the Morlet mother wavelet $\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-t^2/2}$ and parameters $\omega_0 = 6.0, \Delta j = 0.125$ following the Torrence-Compo defaults ([Torrence and Compo, 1998](#)). The band-power in a canonical cycle band is the integral of $|W(s, \tau)|^2$ over scales $s \in [\omega_H^{-1}, \omega_L^{-1}]$. Implementation: PyWavelets `cmor` via `ecowave/cycles/decompose.py:morlet_wavelet`.

Appendix B.3. Hamilton-style Markov-switching AR(1)

The two-regime hidden-Markov AR(1) model

$$y_t = \mu_{s_t} + \phi_{s_t}(y_{t-1} - \mu_{s_t}) + \sigma_{s_t} \epsilon_t, \quad s_t \in \{0, 1\}, \quad \Pr(s_t = j | s_{t-1} = i) = P_{ij}, \quad (\text{B.3})$$

following [Hamilton \(1989\)](#) is fitted by EM (Baum-Welch) on the log-differenced series. The dwell-time of each regime is compared to the canonical cycle band; a vote at Gate 2 is cast for the band whose period brackets the median dwell-time. Implementation: `ecowave/waves/model_e_markov.py` (built on `statsmodels.tsa.regime_switching.MarkovAutoregression`). Test: `tests/test_model_e_markov.py`.

Appendix B.4. PELT changepoint detection

The PELT algorithm ([Killick et al., 2012](#)) solves the penalised optimisation

$$\{\hat{\tau}_k\} = \arg \min_{\{\tau_k\}} \sum_{k=0}^K \mathcal{C}(y_{\tau_k+1:\tau_{k+1}}) + \beta K, \quad (\text{B.4})$$

with $\mathcal{C}$ the L2 segment-cost and β the BIC penalty. The detected changepoints are converted to dwell-times and a Gate 2 vote is cast for the matching cycle band. Implementation: `ecowave/waves/model_d_regime.py` (built on the `ruptures` Python package, `Pelt(model="l2")`). Test: `tests/test_model_d.py`.

Appendix B.5. Bry-Boschan/Harding-Pagan turning points

The classical Bry-Boschan procedure (Bry and Boschan, 1971; Harding and Pagan, 2002) identifies peaks and troughs by a censored local-extremum rule. We adopt the Harding-Pagan minimum-phase duration of five quarters (or two years on annual data) and the minimum-cycle duration of fifteen months. Implementation: `ecowave/waves/model_g_bryboschan.py:_detect_turning_points`. Test: `tests/test_model_g.py`.

Appendix B.6. Dual-null surrogate test

Null 1: AR(1) bootstrap..

- 1: Fit $\hat{\phi}, \hat{\mu}, \hat{\sigma}$ on y by OLS.
- 2: **for** $b = 1, \dots, B = 1000$ **do**
- 3: Draw $\epsilon_t^{(b)} \sim \mathcal{N}(0, \hat{\sigma}^2)$, $t = 1, \dots, T$.
- 4: Simulate $y_t^{(b)} = \hat{\mu} + \hat{\phi}(y_{t-1}^{(b)} - \hat{\mu}) + \epsilon_t^{(b)}$.
- 5: Compute band-power P_b of $y^{(b)}$ on the canonical band.
- 6: **end for**
- 7: Reject Null 1 iff empirical band-power P_0 exceeds the 95th percentile of $\{P_b\}_{b=1}^B$.

Implementation: `ecowave/cycles/surrogate_generators.py:fit_ar1,simulate_ar1`.

Null 2: Theiler phase-randomisation..

- 1: Compute $\hat{y}_k = \mathcal{F}[y]_k$ (DFT, $k = 0, \dots, T - 1$).
- 2: **for** $b = 1, \dots, B = 1000$ **do**
- 3: Draw new phases $\theta_k^{(b)} \sim \text{Uniform}[0, 2\pi)$ independently for $k = 1, \dots, T/2$,
 set $\theta_0^{(b)} = 0$ and impose conjugate symmetry $\theta_{T-k}^{(b)} = -\theta_k^{(b)}$.
- 4: Compute $y^{(b)} = \mathcal{F}^{-1} \left[|\hat{y}_k| e^{i\theta_k^{(b)}} \right]$ (real by construction).
- 5: Compute band-power P_b of $y^{(b)}$ on the canonical band.
- 6: **end for**
- 7: Reject Null 2 iff P_0 exceeds the 95th percentile of $\{P_b\}$.

The conjugate-symmetry constraint (Theiler et al., 1992) ensures real-valued surrogates. Implementation: `ecowave/cycles/surrogate.py:phase_scramble_null`.

Combined dual null. The cell-level p -value is $p_{\text{dual}} = \max(p_1, p_2)$ where p_1, p_2 are the empirical p -values under Null 1 and Null 2. A cell is admitted at Gate 1 iff $p_{\text{dual}} < 0.05$. Implementation: `ecowave/cycles/surrogate.py:dual_null`. Test: `tests/test_surrogate_generators.py` (reproducibility of AR(1) simulation seed; coverage of phase-scramble at B values $\{100, 500, 1000\}$).

Appendix B.7. Calibration of the protocol on synthetic data

The protocol's type-I error and power are calibrated on synthetic panels.

(i) On pure AR(1) panels with $\phi \in \{0.3, 0.6, 0.9\}$ and $T \in \{60, 150, 300\}$, the empirical Gate 1 false positive rate is within two standard errors of 0.05 over $N = 1000$ replicates. (ii) On AR(1) panels with an embedded cosine at the canonical Kondratieff period (50y) and signal-to-noise ratio 1.0, the empirical power is 0.83, above the pre-registered 0.8 threshold. (iii) On the original NBER business-cycle dating chronology, the Bry-Boschan estimator reproduces the published turning points exactly. The calibration report is in

reports/calibration_v1.json.

Appendix C. Pre-registered acceptance criteria

This appendix records, verbatim, the three-gate acceptance protocol that was committed to the project repository before any data ingestion. The protocol is also versioned in `methodology/trois_portes.md` and frozen in code through the immutable `MappingProxyType` export of `ecowave/cycles/bands.py:CYCLE_BANDS`.

Appendix C.1. Falsifiability statement

A cycle is declared *present* on a (cycle, panel, variable) cell iff the cell passes Gates 1, 2 and 3. A cycle is declared *absent* on a cell iff any of the three gates is failed. The verdict is reported on a four-level scale: A (universal), B (regional), C (disputed), D (rejected).

Appendix C.2. Bands, frozen ex ante

Cycle	Band (years)	Anchor references
Kitchin	3–5	Kitchin (1923) ; Wen (2005) ; Diebolt and Doliger (2008)
Juglar	7–11	Juglar (1862) ; Schumpeter (1939)
Kuznets	15–25	Kuznets (1930) ; Korotayev and Tsirel (2010)
Kondratieff	40–60	Kondratieff (1925) ; Modelske (1987) ; Korotayev and Tsirel (2010)

The bands are exported through `MappingProxyType` so they cannot be mutated by downstream code; a unit test (`tests/test_bands_immutable.py`) asserts immutability.

Appendix C.3. Gate 1: dual-null existence

At significance level $\alpha = 0.05$ on $B = 1000$ surrogates per null, the empirical band-power must exceed the 95th percentile of *both* (a) the AR(1)-bootstrap null and (b) the Theiler phase-randomised null. Formally, with P_0 the empirical band-power and $\{P_b^{(1)}\}_{b=1}^B, \{P_b^{(2)}\}_{b=1}^B$ the two surrogate samples,

$$p_{\text{dual}} = \max\left(\frac{1 + \#\{b : P_b^{(1)} \geq P_0\}}{1 + B}, \frac{1 + \#\{b : P_b^{(2)} \geq P_0\}}{1 + B}\right) < 0.05. \quad (\text{C.1})$$

Note the conservative $(B + 1)$ denominator and $+1$ numerator recommended by [Schreiber and Schmitz \(2000\)](#).

Appendix C.4. Gate 2: method consensus

A cell passes Gate 2 iff at least three of the four estimators (Christiano-Fitzgerald, Morlet, Markov-switching, PELT) cast the same phase label (or the matching dwell-band) at every common time point. Pre-registered relaxations: two-of-two for Kitchen on annual data and two-of-three for Kondratieff on the World Bank panel (whose 65-year window admits one full PELT changepoint at most).

Appendix C.5. Gate 3: cross-aggregate universality

A cycle passes Gate 3 iff at least four of the five income aggregates of the World Bank panel (WLD, HIC, UMC, LMC, LIC) independently pass Gates 1–2. A cycle that passes on fewer than four aggregates is recorded as *regional*; a cycle that passes on three or fewer is recorded as *rejected at universality*.

Appendix C.6. Multiplicity

We do *not* apply Bonferroni or Benjamini-Hochberg corrections across the (cycle, panel, variable) grid. The cascade structure provides multiplicity protection by construction: under the joint null (no cycle), the probability of passing all three gates on a single cell is bounded above by $\alpha \cdot p_2 \cdot p_3$, where $p_2 \leq \binom{4}{3}\alpha^2$ (three-of-four consensus under independent estimator nulls) and $p_3 \leq \binom{5}{4}\alpha^4$ (four-of-five aggregate agreement under panel independence). The order-of-magnitude joint type-I error per cell is thus $\sim \alpha^7$, making a Bonferroni correction unnecessary.

Appendix C.7. Per-variable safeguard

A composite-positive, component-negative outcome is recorded as a compositing artefact and does not contribute to the verdict. Formally, a cycle passes the safeguard iff

$$\frac{\#\{v \in \mathcal{V}_{\text{composite}} : p_{\text{dual}}^{(v)} < 0.05\}}{|\mathcal{V}_{\text{composite}}|} \geq 0.5, \quad (\text{C.2})$$

where $\mathcal{V}_{\text{composite}}$ is the set of component variables that enter the composite aggregate. The threshold 0.5 reproduces the convention of [Wen \(2005\)](#): a cycle that is present on the composite but absent on more than half its components is rejected as a compositing artefact.

Appendix C.8. Conditions that would falsify the protocol itself

The protocol is a conjecture. We declare it falsified on observation of any of the following:

- (i) False-positive rate at Gate 1 on synthetic AR(1) panels exceeds $\alpha = 0.05$ by more than two standard errors over $N = 1000$ replicates.

- (ii) Power to detect at Gate 1 on synthetic panels with an embedded cosine at the canonical period (signal-to-noise ratio 1.0) falls below 0.8.
- (iii) Any of the four Gate 2 estimators fails to reproduce its canonical published result on the corresponding textbook benchmark (Bry-Boschan: NBER business-cycle dating; Hamilton: US industrial production 1947–1995; PELT: AR-coal-output benchmark of Killick et al. (2012); Christiano-Fitzgerald: monthly US industrial production (Christiano and Fitzgerald, 2003)).

All three conditions are routinely checked by the project’s test-suite (`tests/`); calibration results are exported to `reports/calibration_v1.json`.

Appendix C.9. Commit fingerprint

The protocol was committed to the project Git repository at commit `<commit-hash>` on `<date>`².

Appendix D. Per-variable evidence and the Wen (2005) replication

This appendix reports the per-variable Gate 1 evidence that underpins Section 5. It demonstrates that the apparent positive results on composite aggregates are uniformly diagnosed as compositing artefacts by the per-variable safeguard (Section 4.1). The replication of Wen (2005)’s inventory-cycle test on the sectoral panel is reported in detail.

²The placeholders are filled at build time by the Makefile target `make pdf` through the variable `COMMIT_HASH`, ensuring that the typeset PDF carries the exact Git fingerprint of the protocol as applied to the data.

Appendix D.1. The sectoral panel: zero of forty-four cells survive

Following [Wen \(2005\)](#), we apply Gate 1 (dual null, $\alpha = 0.05$, $B = 1000$) to every variable of the sectoral panel on each of the four canonical bands.

Table D.3: [Wen \(2005\)](#) replication on the sectoral panel (eleven historical series $\times$ four canonical bands). p_{dual} is the dual-null p -value of Section 3.3; a cell with $p_{\text{dual}} < 0.05$ is recorded as $\checkmark$ (cycle present at Gate 1), otherwise $\times$ (rejected).

Variable (panel period)	Kitchen	Juglar	Kuznets	Kondratieff
Rail freight US (1830–2020)	$\times$	$\times$	$\times$	$\times$
Pig iron production UK (1750–2020)	$\times$	$\times$	$\times$	$\times$
Pig iron production US (1810–2020)	$\times$	$\times$	$\times$	$\times$
Coal output world (1820–2020)	$\times$	$\times$	$\times$	$\times$
Coal output UK (1700–2020)	$\times$	$\times$	$\times$	$\times$
Cotton output US (1790–2020)	$\times$	$\times$	$\times$	$\times$
Wheat output US (1840–2020)	$\times$	$\times$	$\times$	$\times$
Wheat output UK (1700–2020)	$\times$	$\times$	$\times$	$\times$
Industrial production US (1900–2020)	$\times$	$\times$	$\times$	$\times$
Industrial production UK (1750–2020)	$\times$	$\times$	$\times$	$\times$
Industrial production DE (1850–2020)	$\times$	$\times$	$\times$	$\times$
Survival rate	0/11	0/11	0/11	0/11

Zero of forty-four cells survive Gate 1 at $\alpha = 0.05$. The same result obtains on the alternative window 1900–2020 (which removes the pre-industrial volatility), demonstrating that the rejection is not a small-sample artefact of the early historical series. The replication confirms the central diagnosis of [Wen \(2005\)](#): cycles that the composite aggregates appear to support are not visible on the underlying sectoral series.

Appendix D.2. Kondratieff on Bank of England Millennium

We apply Gate 1 to every individual variable of the Bank of England Millennium panel (≈ 130 UK annual series, 1700–2016) on the Kondratieff band. The result is summarised in Table D.4 (top three and bottom three rows by p_{dual}).

Table D.4: Per-variable Kondratieff test on the Bank of England Millennium panel.

Variable	p_{dual}	Gate 1
Real GDP per capita (1700–2016)	0.341	×
Bank Rate (1694–2016)	0.182	×
CPI inflation (1700–2016)	0.471	×
Domestic credit / GDP (1880–2016)	0.225	×
Real residential property (1845–2016)	0.388	×
Unemployment rate (1855–2016)	0.554	×
$\vdots$		
Survival rate	0/ ≈ 130	

No individual UK series survives Gate 1 on the Kondratieff band. The composite-level positive result on the World Bank panel (Section 5) therefore arises from aggregation, not from any underlying universal periodicity. This is consistent with Garvy (1943), Solomou (1987) and Maddison (1991).

Appendix D.3. Kondratieff on Jordà-Schularick-Taylor R6

On the JST R6 panel (18 economies $\times$ 45 variables = 810 cells), the Kondratieff band survives Gate 1 on a tiny minority of cells: 14 of 810, all of which fail the per-variable safeguard when re-tested under the alternative

window 1870–2007 (removing the post-2008 trend). Detailed numbers are exported to `reports/jst_kondratieff_per_variable.json`.

Appendix D.4. Composite vs. component: a diagnostic plot

In each panel where the composite passes Gate 1 but the components do not, we report the same finding diagnostically: the composite’s apparent rejection of the null is an artefact of aggregation. The diagnostic plot is reported in `figures/composite_vs_component.pdf` (generated from `reports/composite_vs_component_v1.json`).